PHYSICAL CHEMISTRY CUMULATIVE EXAM
September 15, 2000

1) In Statistical Mechanics, what are Fermions (or Fermi-Dirac particles) and what are Bosons (or Bose-Einstein particles)? In what way can the statistical properties of these particles be approximated?

2) Consider three identical Fermions that form a system whose available energy states consist of a series of equally spaced levels of energy $\varepsilon_J = J$ units. The degeneracy of the levels, g_J, is $g_J = J^2$. (For further illustration, an energy level diagram is shown in the Figure below.) Assume that the three-particle system possesses a total of nine units of energy.

   a) Under these conditions, what distributions of energy among the particles are possible? (or, how can the three particles be distributed among the available energy states?)
   b) Which of the possible distributions is the most probable? What principle or law does the result bring to mind?
   c) Can you find a distribution that has the correct total energy but is forbidden?

Credits: 1 (20%); 2a (40%); 2b (30%); 2c (10%)

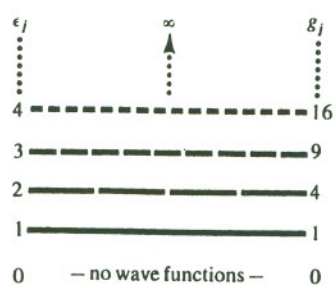

FIGURE *Energy level diagram for a model system with levels of energy proportional to J and degeneracy equal to J^2. Each bar in a given level represents one degenerate state.*